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SODIUM VELOCITY FIELDS IN COMET 1965f*

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ABSTRACT

High-dispersion spectra of the sun-grazing Comet 1965f (Ikeya-Seki) near perihelion show a wealth of detail in the D_1 and D_2 resonance emission lines of sodium. Interpretations given to the main features are: (1) an ejection of Na I toward the Sun over the 4″ dusty nucleus of the comet, (2) the blowback of sodium toward the comet's tail due to the force of radiation pressure on the neutral sodium, and (3) a surprisingly rapid ionization of sodium atoms (a very approximate 50-sec lifetime is derived from velocity profiles in D_2, compared to a predicted *photo-ionization* time of around 83 sec) which limits the maximum blowback velocity to about 30 km sec^{-1}. The destruction of neutral sodium is likely to be due mainly to collisional ionization by electron impact with coronal electrons and photo-ionization by solar ultraviolet radiation.

Observations of the velocity profiles in the D_2-line at different locations in the comet's coma and tail have been compared with simple optically thin models of the Na I velocity field. The agreement is fairly good. However, the reason for simple exponential-decay form of the velocity profiles is obscure A strict cutoff of the maximum blowback velocity is observed on all plates. Some significant optical-depth effects seem present behind the comet's nucleus—this is consistent with the low D_2/D_1 ratio observed by Evans and Malville and Boyce and Sinton.

The presence of low-velocity neutral sodium far from the nucleus implies that most of the sodium is ejected from the nucleus in another form, either as part of a dust matrix or particle or perhaps bound in a parent molecule such as NaCl or NaOH.

The precursor comet-head grains hypothesized are approximately 20 μ in radius, or larger, so that the effect of radiation pressure upon them is compatible with the velocity of sodium in the tail. The dust-continuum distribution has a sharp gradient on the sunward side and falls off slowly toward the comet's tail.

We suggest, as a typical sequence of events, that a parent grain or molecule containing sodium is ejected from the nucleus, and may travel some distance ($\sim 10^4$–10^5 km), then is destroyed and releases Na I atoms. The unscreened neutral sodium is rapidly accelerated away from the Sun into the comet's tail, where it is ionized by an energetic coronal electron or ultraviolet photon some 50 sec later.

A D-line reversal which extends toward the tail is probably the geometric shadow of the dusty nucleus. The sodium atoms in the tail do not "see" the full intensity of the Sun.

I. OBSERVATIONS AND INTRODUCTORY REMARKS

The high-dispersion spectrograms of Comet 1965f (Ikeya-Seki) we discuss in this paper were obtained on October 20 and 21, 1965, (U.T.) with the McMath Solar Telescope at Kitt Peak National Observatory. The linear dispersion of the plates is almost exactly 5 mm Å^{-1} at the sodium D-lines. The exposures were of 30–40-sec duration on III-F and 103aD emulsions. Some of the post-perihelion plates (October 21, 1965) had some additional exposure to the day sky superposed on the comet emission.

The large velocity range in the D-lines was recognized immediately by visual reconnaissance of the spectrum in the early afternoon of October 20. Plates of 5 mm Å^{-1} were then obtained; they cover an interval of about $1^{h}.4$, during which time the comet was only

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a few degrees from the Sun. Thus, the spectrograms are all contaminated with the scattered-light solar spectrum which serves as a very useful wavelength system. Table 1 presents a journal of the pre-perihelion observations of Comet 1965f.

Spectrograms were secured with the entrance slit at several position angles with respect to the apparent radius vector, comet to Sun. A schematic photograph of the slit orientations is shown in Figure 12 (Plate 6). A major problem was that of manually guiding the telescope so that the comet remained fixed on the entrance slit through a given exposure of 30 sec. The angular motion of Comet 1965f near perihelion was very great. Most plates have guiding errors of 2–3″, but some seem better—possibly they have a resolution of 1″. The seeing was very good for the October 20 pre-perihelion observations; it was much worse on October 21. Therefore, most of our discussion that depends upon high geometrical resolution in the comet's head will be restricted to the

TABLE 1

PRE-PERIHELION COMET OBSERVATIONAL DATA AND RADIATION PRESSURE ACCELERATIONS FOR Na I

Plate (IS)	Time (M S T.)	(U.T)	R(a u)	$\dot{\rho}$(km sec^{-1})	$\Delta\lambda D_2$(Å)	b_r (km sec^{-2})
9	14:13	21:13	0 0427	+110 95	+2 178	0 27
10 .	14:13	21:13	.0427	110 95	2 178	27
11 .	14:24	21:24	0419	112 89	2 216	.27
12	14:27	21:27	.0417	113 43	2 227	28
13 .	14:28	21:28	0416	113 62	2 231	28
14	14:28	21:28	.0416	113 62	2 231	.28
17	14:54	21:54	.0397	118 60	2 329	31
18	14:54	21:54	0397	118 60	2 329	31
19	14:56	21:56	0395	119 01	2 337	32
20	14:58	21:58	0394	119 42	2 345	.33
21	15:06	22:06	0388	121 09	2 377	33
22	15:08	22:08	0386	121 51	2 386	33
25	15:32	22:32	0367	126 92	2 492	.36
26	15:35	22:35	0 0365	+127 63	+2 506	0 37

pre-perihelion observations of October 20, 1965. The scale of the solar telescope is 2″.26 mm^{-1}; our best plates can show structure of about 0.5 mm or details of ~700 km at the comet, for the position angles in which foreshortening plays a minor role.

As mentioned in a preliminary report (Livingston *et al.* 1966) a most prominent feature of these pre-perihelion spectrograms is the sharp red edge to the D-lines, especially at the nucleus of Comet 1965f. There are rapid changes in plate density, both with wavelength (velocity) and with position in the head and coma, which have necessitated several modes of presentation. For ease of interpretation, microphotometer tracings and isophotes of the spectra (to be described later) are more instructive. The instrumental velocity resolution is about 1 km sec^{-1}.

II. EJECTION OF SODIUM FROM THE HEAD OF THE COMET

One of the most striking features of the pre-perihelion D-line spectra is the sharp redward edge of the D profiles, especially over the dusty reflected-light continuum strip produced by the comet's nucleus. See Figures 1 and 2 (Plates 1 and 2).

At the nucleus, the D-lines bulge toward longer wavelengths by an amount corresponding to a ***maximum velocity*** toward the Sun of 5 km sec^{-1} (taking projection effects into account). The bulge is ***strictly limited*** to the 4″-diameter nucleus. On every plate taken at any slit orientation, the strict confinement of the redward bulge to the nucleus

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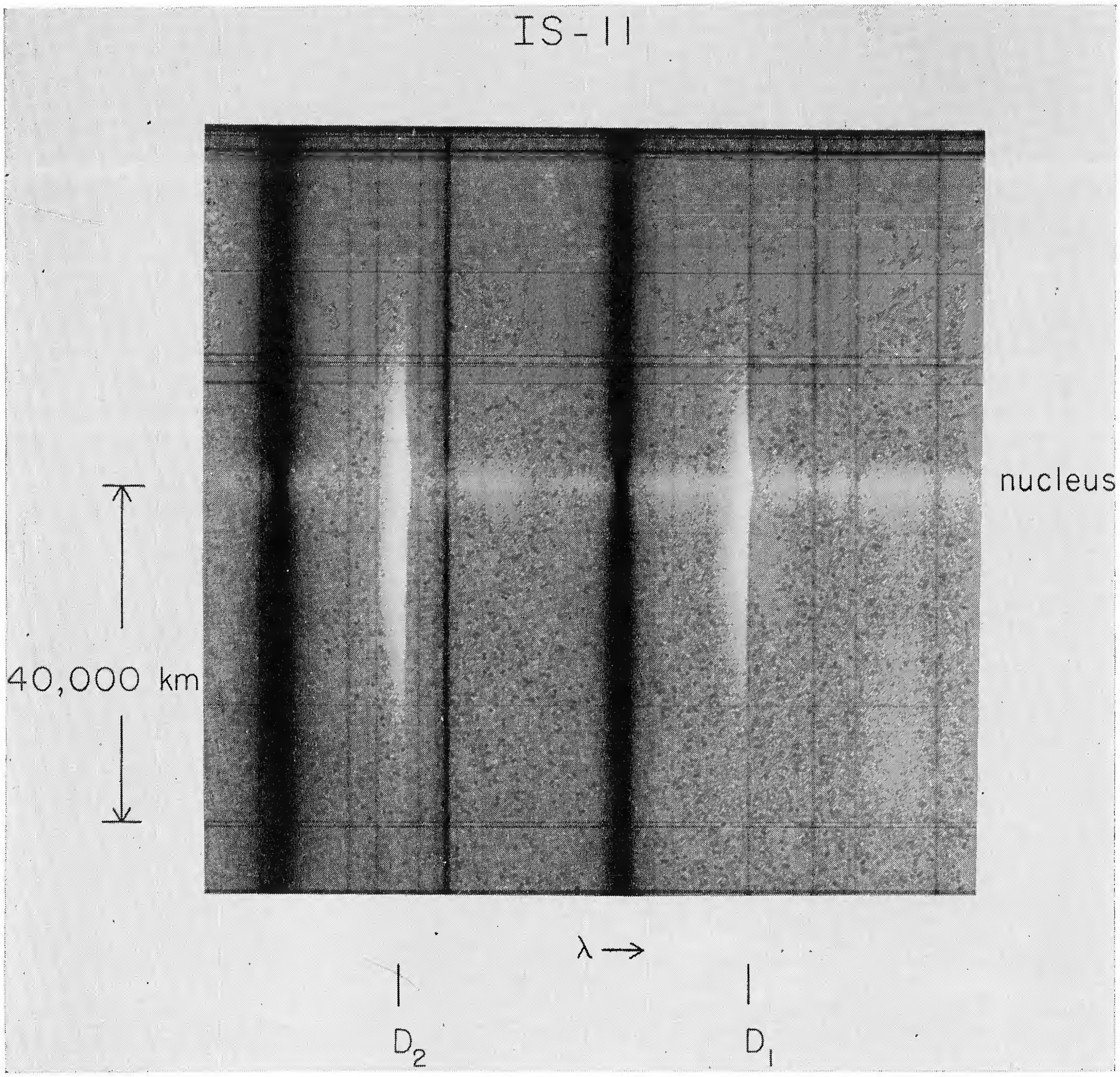

FIG. 1.—The sodium D-lines in Comet 1965f on plate IS 11. The strong absorption lines in the background are from the day-sky spectrum near the Sun. The continuum streak in the middle is due to the comet's nucleus. On this spectrogram the slit was oriented approximately at right angles to the comet-Sun line, so that we see N I emission in the coma of Comet Ikeya-Seki to about 40000 km from the head. Note the D-line bulge to the red (velocity away from Earth, toward the Sun) over the nucleus. The extensions of the emission lines into the coma also have a well-defined red edge. The original dispersion was 5 mm Å^{-1}.

SPINRAD AND MINER (*see* page 356)

PLATE 2

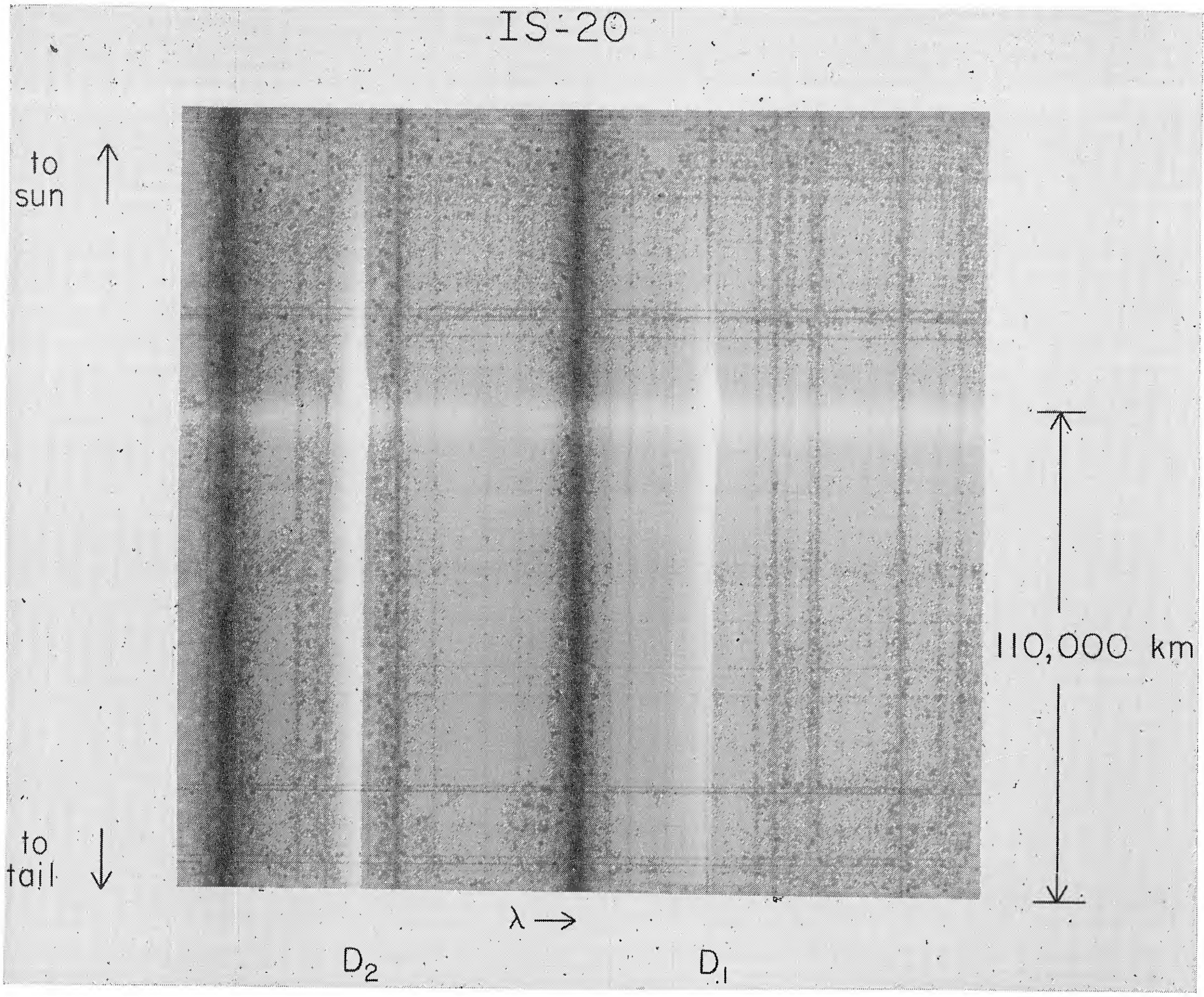

FIG. 2.—The D-lines of Comet 1965f on plate IS 20, where the slit orientation was along the Sun-comet radius vector. Note the strong Na I emission in the near tail, the sharp redward edges of D_1 and D_2, and again the nuclear bulge of the emission lines to the red. The latter phenomenon probably represents the ejection of neutral sodium toward the Sun. On this plate the D_2-line extends well beyond 110000 km in the tail of the comet (taking foreshortening into account), yet the Na I velocity in the tail is quite low.

SPINRAD AND MINER (*see* page 356)

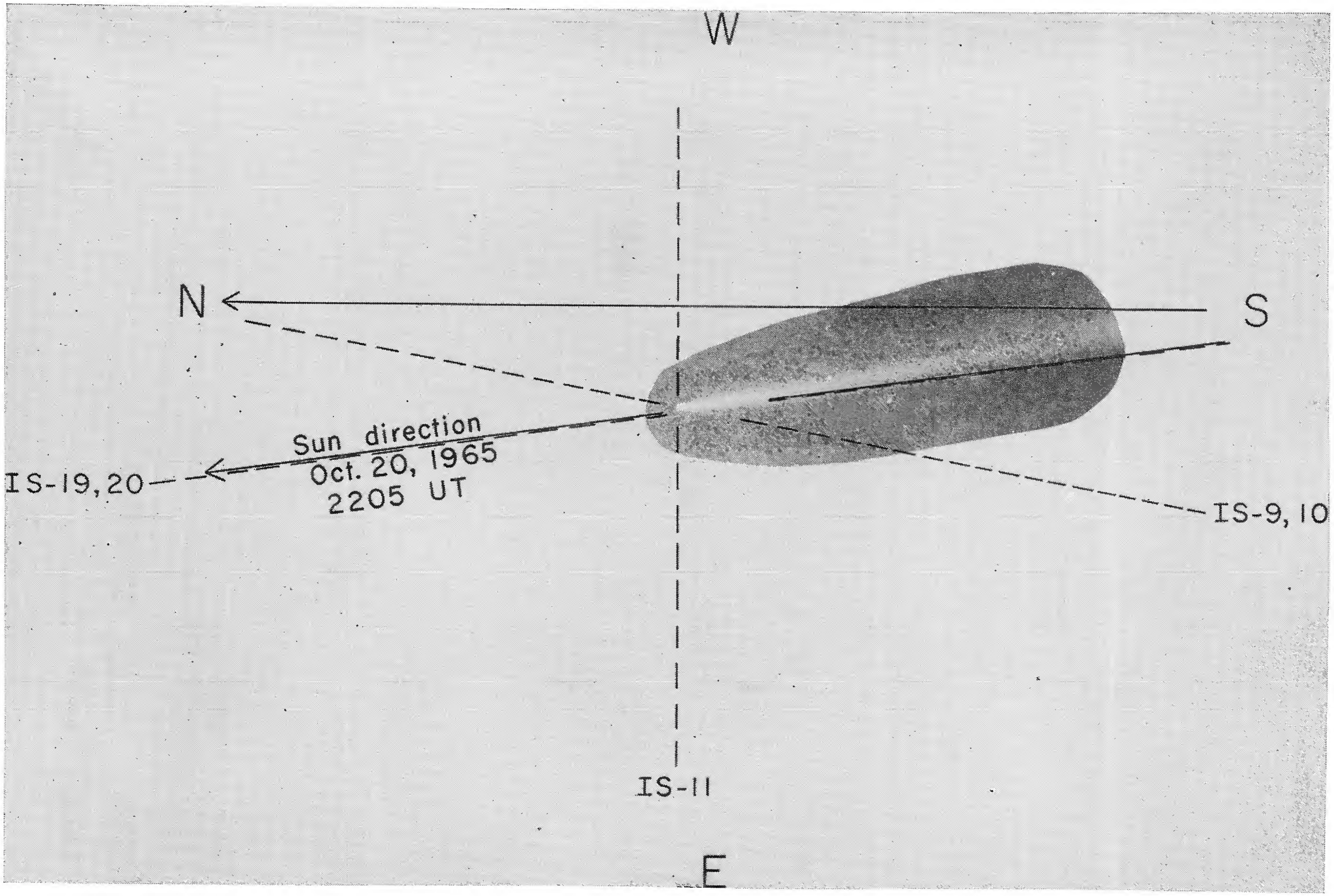

FIG. 12.—The approximate angular orientations of the spectrograph slit for some of the spectrograms of Comet Ikeya-Seki discussed here. The Sun-comet direction indicated pertains to October 20, 1965, at 2205 U.T., or near the end of this KPNO series of plates. The position angle of the comet's tail varied over the $1^h.5$ interval of exposures, so the dotted-line slit orientations are not equally precise. The photograph of Comet 1965f superposed is a post-perihelion exposure by Mr. C. Capen at Table Mountain Observatory. The tail of the comet was not nearly this spectacular to the eye on October 20, 1965.

SPINRAD AND MINER (*see* page 356)

and the maximum velocity toward the Sun of 5 km sec^{-1} is maintained. On the poorer post-perihelion spectrograms, the sharp edge is on the violet side, i.e., still directed toward the Sun in velocity. The bulge is less obvious on these plates.

There is no indication of a comparable ejection of Na I away from the Sun at the nucleus or elsewhere. The *mean ejection velocity* of about 3.0 km sec^{-1} directed toward the Sun is interpreted to be the actual ejection velocity of Na I (and probably other metallic vapors) from the comet's nucleus. This may be a sign of chemical explosions on the solar-heated side of the nucleus of Comet 1965f.

At the edges of the dusty nucleus the positive radial velocity is reduced, but we cannot say whether this is a wholly physical phenomenon in the comet or partly an effect of the geometry, since radial ejection of Na I from the comet's nucleus at 90° to the solar radius vector would be in the transverse direction as seen from Earth, for our pre-perihelion observations. Thus, the ejection might take place radially over the whole hemisphere illuminated by sunlight. The effect is quite noticeable on Figures 3 and 4 (Plates 3 and 4), which are typical isophotes of representative spectra. A consequence of the sunward directionality of the initial ejection of sodium is the presence of Na I on the Sun side of the nucleus, and indeed on spectrograms IS 19, 20, 25, and 26 with the slit orientation along the radius vector to the Sun, D emission extends as much as 6 mm or some 16000 km (taking projection effects into account) to the Sun side of the nucleus. If it were not for the strong influence of solar radiation pressure on the Na I atoms and their probable grain precursors, the sunward extension would be even greater. A somewhat similar phenomenon was observed by Greenstein and Arpigny (1962) in Comet Mrkos (1957d). We discuss radiation pressure and Na I destruction next.

III. RADIATION PRESSURE BLOWBACK AND SODIUM IONIZATION

At a solar distance of $R = 0.04$ a.u. the force of repulsion on the neutral sodium atoms in Comet 1965f due to radiation pressure was quite large. Since the heliocentric radial velocity, dR/dt, was large (-187 km sec^{-1}), the comet's D_1 and D_2 resonance rest wavelengths were illuminated by the full solar continuum near λ5890. The Ni I line at λ5893 played no substantial role in reducing the solar flux either. Then, following Wurm's (1963) formula, we list in the last column of Table 1 the antisolar accelerations (b_r) on Na I atoms due to radiation pressure at the times of interest. Typical accelerations are $b_r \sim 0.3$ km sec^{-2}. These apply, of course, to regions of the comet which are optically thin in D_1 and D_2 light. In our situation, the tabulated b_r values should be accurate accelerations for neutral sodium atoms between the nucleus of the comet and the Sun, and for atoms far out in the tail. The computed accelerations might be overestimates for the bright coma and inner tail of the comet when the Na I density is high, and the optical depth in D could be considerable.

If Na I remained exposed to sunlight in this environment for a considerable length of time, say for some 300 sec, then we should observe Na I radial velocities up to $\sim$90 km sec^{-1} on our spectrograms. Since these high velocities are *not* seen, some process must quickly destroy Na I. Ionization of neutral sodium is the most obvious and reasonable way to accomplish destruction; we consider *photo-ionization* and *collisional ionization* as methods of destroying the neutral sodium atoms.

Wurm (1963) calculated the lifetime of Na I atoms against solar photo-ionization, assuming that the ionization continuum below the 5.14 eV threshold was hydrogenic [$a_\nu \sim (\nu/\nu_0)^{-3}$]. However this result is incorrect; the photoabsorption coefficient is much smaller than calculated by Wurm. Modern laboratory experimental results by Hudson (1964) on the Na I cross-section just below the threshold of ionization are now available; we have computed the probability of Na I photo-ionization by folding the cross-sections into the solar fluxes tabulated by Tousey (1963) and integrating. The most important contribution to the probability of ionization is from ultraviolet light between λλ2400–2300.

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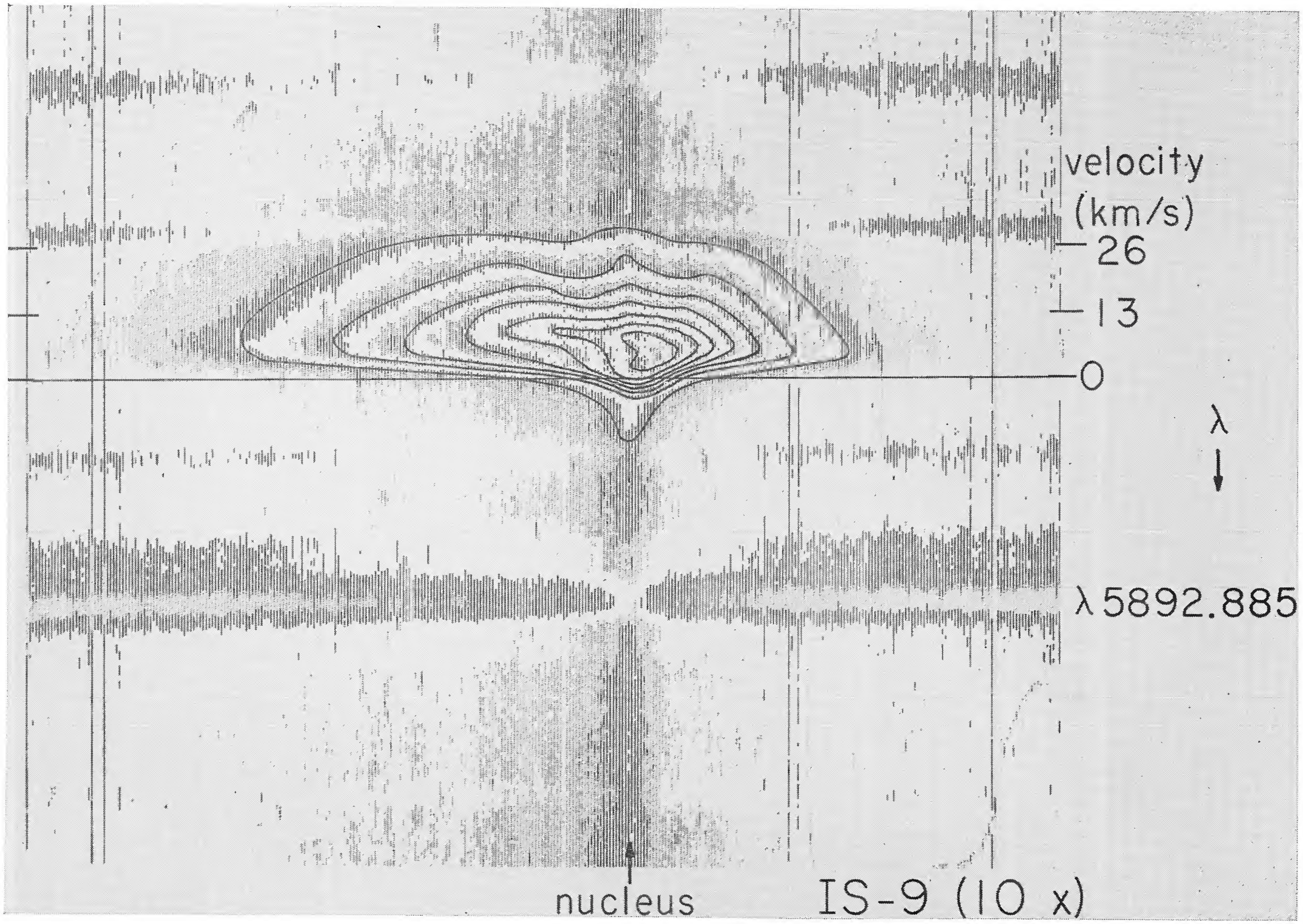

FIG. 3.—Equal-density isophotes of the D_2-line on plate IS 9 with 10× magnification. The velocity scale is indicated; the zero point (systemic comet velocity) was determined from Cunningham's ephemeris. The slit orientation was (nearly north-south, or some 30°) to the Sun-comet radius vector, so that the isophotes are fairly symmetric with respect to the central continuum strip from the dusty nucleus. The isophotes show the velocity bulge (toward negative velocity in this presentation) over the nucleus (somewhat exaggerated in this isophote) and the relative constancy over the entire coma of the maximum blowback velocity near 25 km sec^{-1}.

SPINRAD AND MINER (*see* page 357)

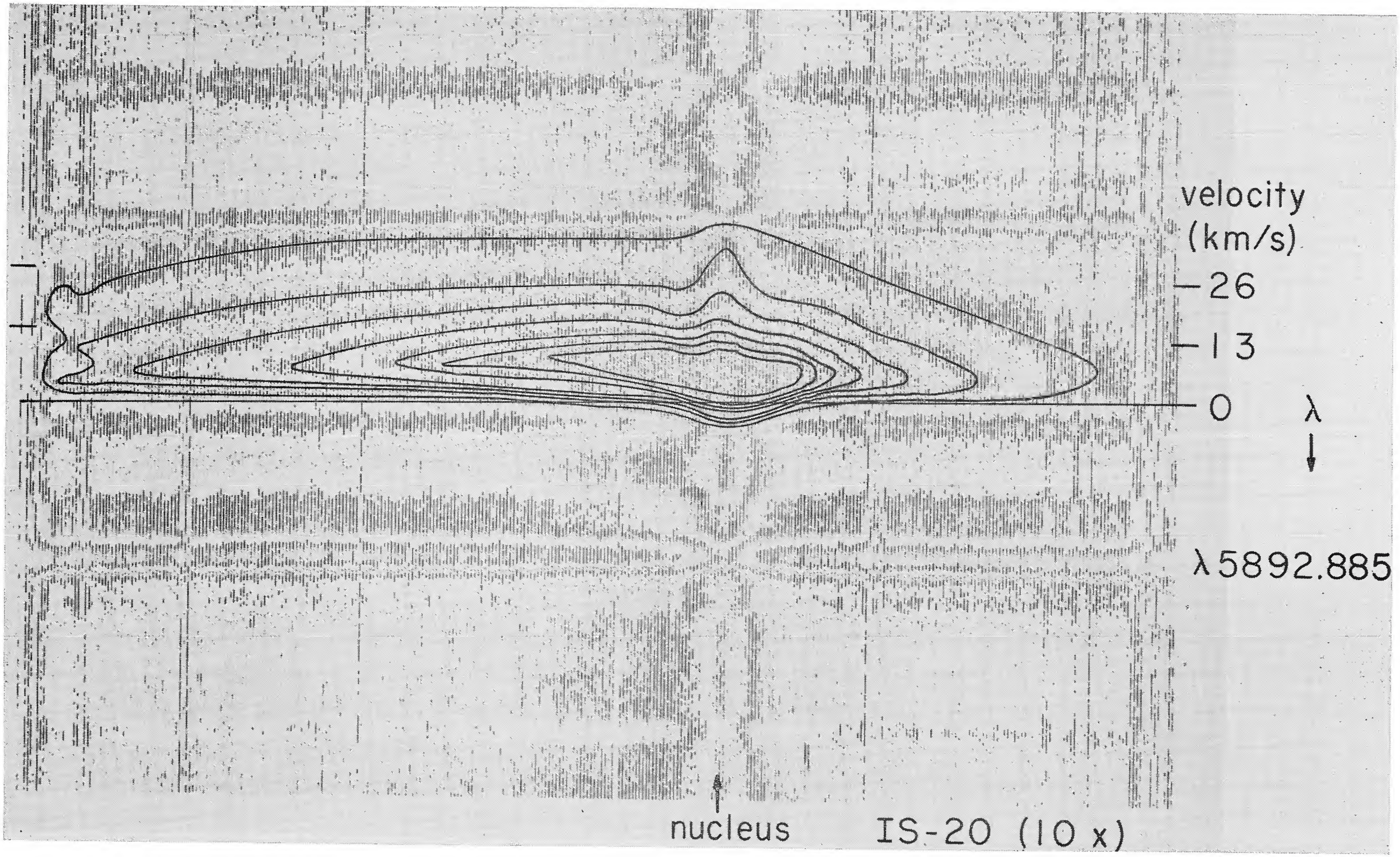

FIG. 4.—Isophotes for IS 20, a spectrogram which was aligned along the tail of Comet 1965f. Besides the velocity bulge toward the Sun (in velocity space) at the nucleus, note the marked asymmetry of Na I emission toward the tail (in space), which is the left-hand side of this figure. The upper isophote here ($\sim$30 km sec^{-1}) is again rather constant over a distance of some 10^5 km in the tail. Note the sloping "leading edges" of the isophotes in the tail; Na I atoms slowly acquire a velocity up to $+3$ km sec^{-1} at the left end of the plate, some 110000 km from the nucleus.

SPINRAD AND MINER (*see* page 357)

We find $\tau_{\text{photo}} = 83$ sec at $R = 0.04$ a.u. The determination is probably accurate to about 10 per cent.

The alternative proposal of collisional ionization is a more difficult hypothesis to check. Internal cometary collisions of sodium atoms with other atomic species, cyanogen molecules, or unobservable molecules (say H_2) are possible, and indeed may occur, but probably on a rather long time scale, probably longer than 200 sec (Preston 1967). Probably not all collisions will ionize the sodium atoms. Moreover, we see from the isophotes in Figures 3 and 4 that the destruction of Na I, as indicated by the *maximum* blowback velocity toward the tail of the comet, is not dependent on the absolute Na I density (and therefore the total particle density) in the comet's coma and tail; the constancy of maximum velocity across the comet implies the collisional ionization agent is *external* to Comet 1965f and therefore probably due to interaction with the solar corona. The typical distance of $R = 0.04$ a.u. is equivalent to $R = 8.6\, R_{\odot}$.

We consider charge-exchange collisions with solar protons first, i.e., $H^+ + Na \rightarrow H + Na^+$. Equation (1) gives the inverse time scale for charge-exchange collisions:

$$\frac{1}{\tau_{\text{charge-exchange}}} = \sigma_{\text{charge-exchange}} V N_p , \tag{1}$$

where V is the proton–Na I velocity and N_p is the coronal proton density at $8.6\, R_{\odot}$. The value of $\sigma_{\text{charge-exchange}}$ given by McDaniel (1964) is $\sim 10^{-16}$ for protons on H and H_2; we arbitrarily guess the same coefficient for sodium. We guess about 4×10^4 for N_p (Parker 1960).

Our result is $1/\tau_{\text{charge-exchange}} = 8 \times 10^{-5}$, probably accurate to only an order of magnitude. Thus $\tau_{\text{charge-exchange}} \approx 10^4$ sec, so slow an ionization rate as to be out of consideration in our context.

However, collisional ionization by electron impact is more favorable. The faster moving electrons increase the probability of Na I ionization. We compute

$$\frac{1}{\tau_{\text{electron impact}}} = N_e \int_{v_{\text{th}}}^{\infty} Q_v v f(v)\, dv \tag{2}$$

(Preston 1967), where N_e is the coronal electron density and the lower limit of integration, v_{th} is the electron velocity corresponding to the 5.1 eV threshold for Na I ionization, v is the space velocity, $f(v)$ is the Maxwellian velocity distribution, here chosen to correspond to that at $T = 10^6\,°$ K. Q_v is the cross-section for ionization by electron impact as given by McFarland and Kinney (1965). We are indebted to V. Myerscough for numerical evaluation of the integrand at a dozen electron energies and the computation for integral in this formula. Our result agrees well with the tabulation by Lotz (1967). We find

$$\frac{1}{\tau_{\text{electron impact}}} = 1.76 \times 10^{-7} N_e , \qquad \text{for} \qquad T = 10^6\,°\,\text{K}. \tag{3}$$

We adopt Parker's (1960) coronal model with a density of $N_e = 4 \times 10^4$ cm^{-3} at $R = 8.6\, R_{\odot}$ and find the lifetime of Na I against collisional ionization of $\tau_{\text{electron impact}} =$ 140 sec. The estimate of the coronal electron density is the weakest link in the chain of argument; N_e might be as much as a factor 3 different from our assumed value, and the collisional destruction time of neutral sodium atoms in the comet is inversely proportional to N_e. If the rest of the data used in this paper were more precise, we could invert the procedure and use the D-line profiles to solve for the coronal electron density.

Since there is a competition between photo- and collisional ionization, we compute the time scale against Na I ionization by adding the reciprocal times:

$$\frac{1}{\tau_{\text{Na}}}=\frac{1}{\tau_{\text{photo}}}+\frac{1}{\tau_{\text{electron impact}}} \tag{4}$$

yielding $\tau_{\text{Na}} = 53$ sec for our predicted sodium-destruction time scale.

We note that once a sodium atom is ionized it will not recombine very quickly here; Na^+ to Na radiative recombination can be considered negligible in this situation (Seaton 1951). Perhaps, as Dr. M. Belton (private communication) mentions, some ionized sodium could become neutral through charge-transfer collisions with other cometary molecules, but this rate of recombination is probably also very slow.

The D_2 emission line (λ5889.977) was traced on a conventional microphotometer, in the density mode, on several of the better pre-perihelion plates of October 20 (U.T.). The velocity profiles for IS 10 and IS 20 are illustrated at several places on the comet (Figs. 5–8), the y-axis being defined at the head (center of dusty continuum streak). Positive y is in the solar direction; negative, toward the comet's tail. All tracings are velocity profiles, i.e., exactly along the direction of the dispersion. The microphotometer slit length, Δy, was always 0.5 mm.

The arrow and zero point of the velocity scale (abscissa) were determined from an orbital ephemeris kindly computed by Dr. L. E. Cunningham at Berkeley for the times in question. The residual uncertainty on the orbit of 1965f near the Sun leads to a possible error up to 1.0 km sec^{-1} in the systemic velocity of the comet and hence in our velocity zero point. This 1 km sec^{-1} velocity uncertainty affects none of our conclusions in a substantial manner.

Isophotometric tracings, or simply isophotes, for all the D_2-line spectrograms, were obtained by Miner with the Jet Propulsion Laboratory isodensity tracer. This method of presentation saved much time, since two-dimensional density pictures of each spectrum were easy to examine for gross details, as the position of the maximum blowback (positive) velocity (note again Figs. 3 and 4, Pls. 3 and 4). However, the finer detail is best illustrated on the one-dimensional tracings.

Unfortunately we do not have a satisfactory intensity calibration for the spectrograms discussed here; the tracings illustrated in Figures 5–8 are uncalibrated, so that the ordinate is plate-blackening, 1-transmission of the spectrogram. The ordinate should be nearly linear in intensity for the more weakly exposed portions of the velocity profiles. The tops of the profiles may not share this characteristic. The consequent distortion will play little role in the sodium velocity model fitting attempts which follow.

Our profile figures have the true cometary velocity scale, 1.25 times the measured radial velocities (because of the projection effect).

On IS 10, a spectrogram taken with the spectrograph slit about 30° to the comet-Sun line, we note the following characteristics of the tracings in Figure 5: first, the bulge toward the sunward velocities is present only on trace $y = 0$ (nucleus), and there is a gradual decay of the emission line to zero intensity at *antisolar* velocity $= +30$ km sec^{-1}. The surprising and significant result is that the maximum velocity away from the Sun, $+30$ km sec^{-1}, is independent of position in the coma and peak density of the cometary Na I emission. A comparison of $y = +10$, $y = -10$ shows this nicely. The difference in slope between $y = +10$ (gradual decay of Na I with velocity ahead of comet on Sun side) compared to the $y = -10$ position (in the tail; sharp drop in Na I with velocity) is probably an indication of optically thick D_2 at $y = -10$. There b_r will be reduced in the "shaded region," while the collisional and radiative ionization will be the same as at $y = +10$. Thus, the radiation pressure will drop, while the lifetime of a neutral sodium atom in the near tail will continue to be about 50 sec. Therefore, screened neutral sodium atoms at $y = -10$, $0 \leq v \leq 10$ km sec^{-1} will acquire little velocity away from the

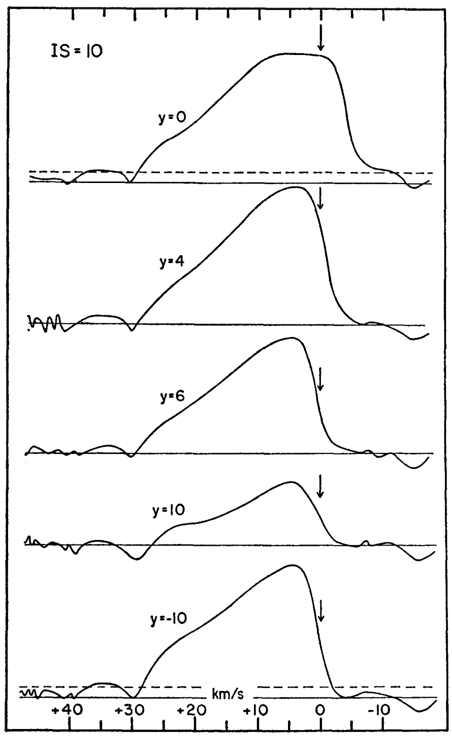

FIG 5

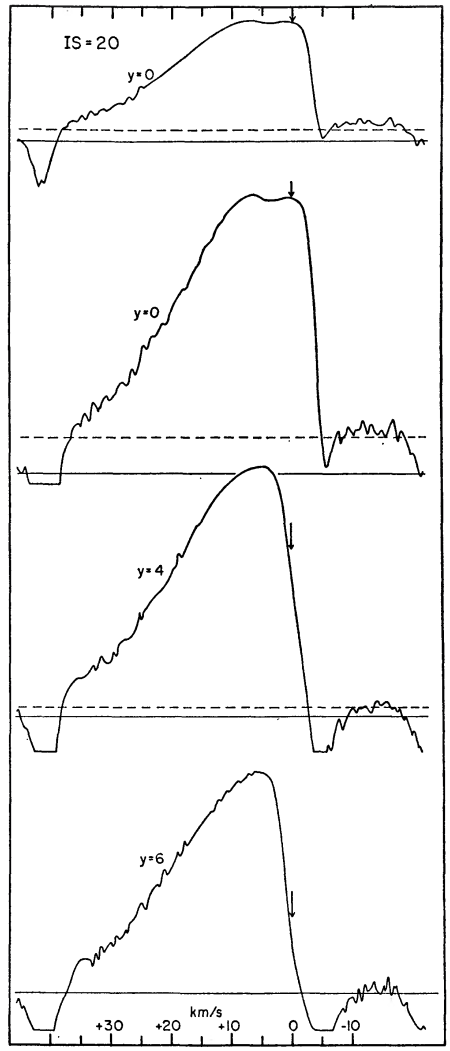

FIG 6

FIG. 5.—Sodium D_2-line profiles, or velocity profiles, traced at different positions perpendicular to the dispersion on plate IS 10. The ordinate is the D_2 plate blackening (roughly proportional to the strength of the Na I emission). The ordinate on each profile is approximately linear in intensity, except at the very top. The velocity zero point, noted by an arrow above the profiles, was established by the orbital ephemeris. The $y = 0$ trace is over the comet's nucleus, and our convention sets positive y in the direction of the Sun. y is measured in millimeters perpendicular to the dispersion of the plage. For the slit position angle of this spectrogram, 1 mm = 1500 km at the comet. Δy, the slit length, was 0.5 mm.

Note the difference in the slope between the lower two velocity profiles at $y = +10$, $y = -10$. This is caused by the substantial optical depth of the D_2-line at the near-tail position $y = -10$. See the discussion in the text. The dotted line is a possible alternate zero-density base line and indicates the maximum uncertainty in this quantity.

FIG. 6.—Velocity profiles from plate IS 20, a plate aligned along the comet-tail–Sun line. The flat top at $y = 0$ is a plate-saturation effect. Two different density magnifications at the nucleus ($y = 0$) are presented. The ordinate for the upper one is non-linear. Note the progressive blowback of the "leading edge" at $v = 0$ km sec^{-1} from $y = 0$ to $y = +6$ on the Sun side.

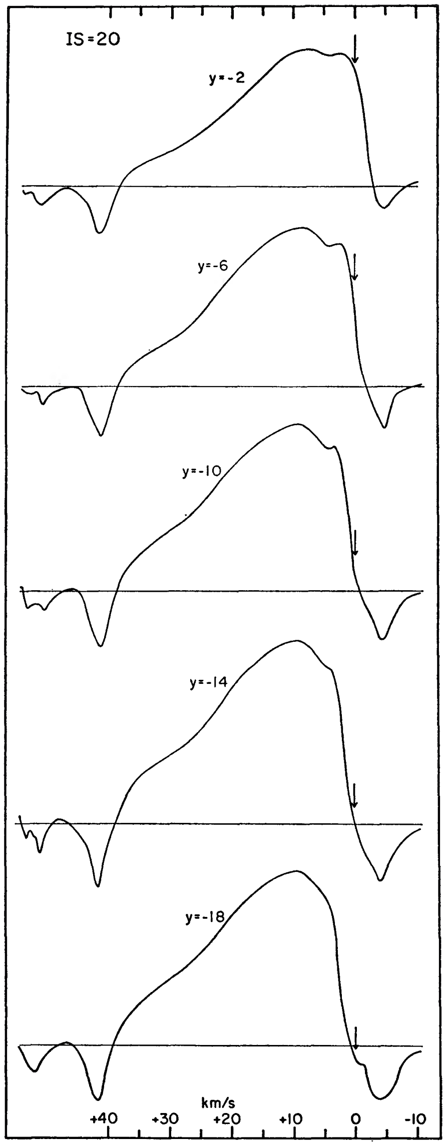

FIG 7

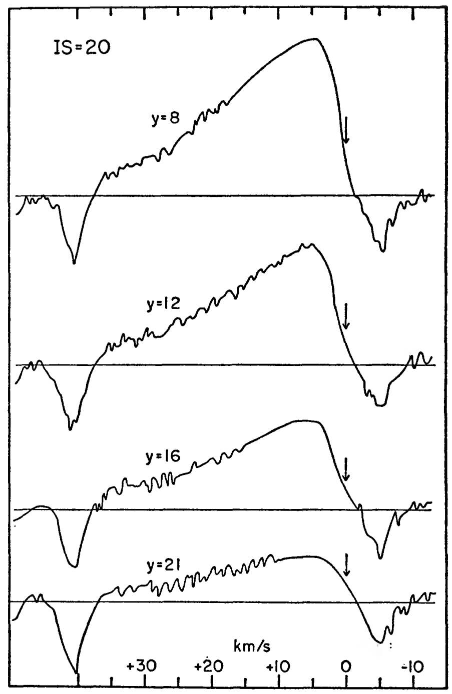

FIG 8

FIG. 7.—More velocity profiles from IS 20. These traces are all on the tail side of the nucleus ($y < 0$). Note the dip at $v = 4$ km sec^{-1} on the upper four traces; it is the reversal discussed in § V of the text. Note also the steep decline of the profiles from $v = 10$ to $v = 40$ km sec^{-1}; this is, in part, due to large optical depth in the D_2-line which reduces the effect of solar radiation pressure on the Na I atoms.

FIG. 8.—Velocity tracings from IS 20 on the Sun side of the nucleus. Here the slope from $v = 10$ to $v = 40$ km sec^{-1} is quite shallow and indicates little shelf-shielding by cometary sodium. Thus, the simple exponential-decay models may be fit to these profiles with their "source" at the density maximum. They yield $\tau_{Na} = 50$ sec for the typical lifetime of a neutral sodium atom. Again, the arrow defining zero velocity indicates the comet's systemic velocity.

comet's nucleus and tend to be ionized before they can recede into the tail rapidly. The velocity profile is steepest in the most shielded, optically thick positions. The suggestion of considerable optical-depth effects at *some* positions in the coma is substantiated by the independent measures of the D_2/D_1 ratio by Evans and Malville (1967), Boyce and Sinton (1966), and Preston (1967). Their line ratios indicate considerable saturation.

Similar optical-depth effects, as on plate IS 10, occur on the tracings of IS 20; they are particularly noticeable in the steep profiles for $y = -6, -10, -14, -18, -21, -24$, and -30 (Figs. 6–8). On plate IS 20 we traced the D_2-line far into the comet's tail ($y = -36$ mm or 10^5 km from the head) and also well ahead of the comet, toward the Sun at a probably unshielded region ($y = 16, 21$). The maximum blowback velocity in each case is about 40 km sec^{-1}, again constant on all the tracings. Note that the b_r acceleration relevant for IS 20 was 20 per cent greater than that for the time of IS 10.

To interpret the shapes of the best tracings ahead of the nucleus in the relatively unshielded optically thin region, we compared IS 20 numbers $y = 16$, $y = 21$, and IS 10 number $y = 10$ to a simple exponential-decay Na I model. The model starts with a low-velocity neutral sodium atom experiencing an acceleration $b_r = 0.3$ km sec^{-2} and having an e-folding ionization time τ as a parameter.

Our mode is a slightly restricted version of the Eddington Fountain model (Eddington 1910) which has also been discussed by Fokker (1953). We limit the velocity of ejection of sodium from the comet's nucleus to low values, $V_{\text{eject}} \leq 4$ km sec^{-1}, from our direct observation of the nuclear velocity bulge, and thus the velocity of the Na I atom acquired through acceleration of radiation pressure will usually dominate over the initial ejection velocity. Physically, our model might be termed a "weak-fountain" model.

We may then compute the form of the velocity distribution to be expected when we look along a narrow column through the comet at a distance y from the nucleus. Dr. M. Belton (private communication) pointed out that we may simply consider the number of sodium atoms traversing a volume element in the line-of-sight column. The density of atoms with velocity between v and $v + dv$ is $N(v)$, and the *rate* at which molecules with the required velocity reach our volume element is $vN(v)$. The probability that a sodium atom will get from its place of origin to the volume element in question is $e^{-v/b_r\tau}$. Let the production rate at the origin of the sodium atoms (as we see later, the place of liberation of atoms from a grain) be dN_s/dt. Then we have

$$vN(v) = \frac{dN_s}{dt}\, e^{-v/b_r\tau}\,. \tag{5}$$

Now if we presume the production rate to be a *constant*, independent of velocity, say N_0, we have

$$\frac{N(v)}{N_0} = \frac{1}{v}\, e^{-v/b_r\tau}\,, \tag{6}$$

which gives the relative number of neutral sodium atoms at any velocity. For comparison with our profiles we set N_0 equal to the maximum Na I 5890 line density on the comet tracing to be examined.

The divergence for $v = 0$ is not a real physical problem, for there will be a dispersion in velocities partly due to a typical initial photodissociation velocity of 2–3 km sec^{-1}.

Unfortunately the observed velocity profiles in the unshielded optically thin portions of the comet, as shown for tracings of plates IS 10 ($y = 6$ and 10) and IS 20 ($y = 12, 16$, and 21), have rather shallow gradients, much more like simple exponential decays. There is no obvious case involving the fast falloff of $N(v)$ dictated by the $1/v$ term in equation (6); this peculiar situation is certainly a puzzling dilemma. Could the source of the sodium atoms eject at preferentially high velocities, so that $dN_s/dt = v$? That would provide a compensation in equation (6). Of course, our simple model may turn out to be inappropriate.

The comparison of the observed velocity profiles with exponential models without the $1/v$ term gives fairly consistent solutions for τ in all cases; the time scale for Na I destruction is about 50 sec with an internal uncertainty of about 5 sec. However, the unsolved dilemma discussed above weakens our confidence in the derived e-folding time; we should consider the decay-time derivation a very approximate solution. Nevertheless, the empirical value of $\tau = 50 \pm 5$ sec agrees well with the τ_{Na} derived theoretically.

IV. A POSSIBLE ORIGIN FOR THE SODIUM

An intriguing problem is brought up by the *mere presence* of Na I at radial distances of some 100000 km in the tail and outer coma of Comet 1965f. If a Na I atom migrated to that distance from the comet's head at a mean velocity of about 10 km sec^{-1}, say, it would take some 10000 sec or $\sim$3 hours for the trip. From previous calculations, we know the Na I cannot remain neutral for long (say $\tau = 50$ sec) and cannot remain at the observed low velocity if, somehow, it succeeded in staying neutral for many minutes. A solution to this problem may be the ejection of sodium in a dust grain or matrix assembly, or in a parent molecule. There is no shortage of rather tightly bound sodium-carrying molecules; we list as examples in order of increasing dissociation energy, D_0, the following sodium compounds: $(Na)_2$, NaH, NaO, NaF, NaCl, NaBr, NaOH, and $Na_2O_2H_2$. We have insufficient information to decide on a reasonable photodissociation time scale for sodium compounds at $R = 0.04$ a.u., but the destruction time for these fairly fragile molecules might be seconds or minutes. However, we need more than 1 hour to transport the sodium compounds to the comet's tail.

Ejection of a dust grain might insure longer stability against destruction by sunlight or collisions with coronal particles, but radiation pressure on a dust grain, assumed to be spherical, is not negligible. Small grains, say with radii $a < 10\ \mu$, will be rapidly accelerated to high velocity and travel to the extreme tail of the comet. We suggest considerably larger dust particles than the normal 1-μ radius, since these will be less subject to radiation pressure (and inadmissible high velocity away from the Sun) and may also survive long enough to release their sodium at large distances from the cometary nucleus. In fact, we may find a rough value for the mean lifetime and size of these parent grains from the observed shape of the Na I line isophotes. The "leading edge" of the isophotes for IS 9 and IS 20, in Figures 3 and 4 (Plates 3 and 4), can be seen to be swept far back into the tail by about a $\Delta v = 4$ km sec^{-1} velocity (say $\Delta x \sim 6 \times 10^4$ km from the head). The slope of the leading edge is then 4 km sec^{-1} 60000 km^{-1} distance.

Assume that the shape of the edge—or the locus of *Na I release points*—is primarily due to radiation forces on the grains. We have two simple equations: one for the velocity as a function of time and radiation-pressure acceleration, and one for the distance traversed toward the comet's tail in the variables acceleration and sodium-liberation time. This simple approach gives us a unique acceleration value for a particular tail distance, Δx. Accordingly, we solve for both the grain acceleration and the sodium-liberation time of the dust. We find a liberation time of $\sim 3 \times 10^4$ sec or about 8 hours for these hypothesized Na I precursors. Their mean acceleration is 13 cm sec^{-2} at 6×10^4 km from the comet's nucleus, which for spherical particles of unit density gives a mean radius of 20 μ (Brandt and Hodge 1964, chap. ix). Our data are consistent with the presence of even higher grains. Our expectation of dust in the comet's tail is substantiated by a series of measures of the radial distribution of continuum light on several spectrograms in this series. On IS 19, in Figure 9, we illustrate a microphotometer tracing at the λ5892 continuum (no emission whatsoever) from the Sun side over the obvious nucleus toward the tail. Note the asymmetry; most of the dust causing the weak reflected-light spectrum has been pushed back a substantial distance (measurable out to

70000 km) on the tail side, but some dust is apparently ejected at the nucleus in the sunward direction, possibly at the observed original Na I ejection velocity of some 3 km sec^{-1}. On Figure 10 we compare the dust and Na I D_1 emission profiles from plate IS 19. The dust peak defining the comet's head is noticeable. The Na I decays at approximately the same rate as the dust continuum in the tail at distances greater than 25000 km from the nucleus. That dust, indeed, is the Na I precursor is weakly indicated by these tracings.

To summarize this section we suggest, as a rather typical series of events in the comet, that a sodium atom bound in a grain (or possibly a tightly bound molecule) is ejected from the comet's nucleus at 3 km sec^{-1} toward the Sun. It may, of course, have a velocity oriented at random with respect to the radius vector also. This sunward-directed velocity is gradually decreased due to radiation pressure on the grain. After a few hours, the grain may have traveled a considerable distance from the nucleus. Then it evaporates, and the

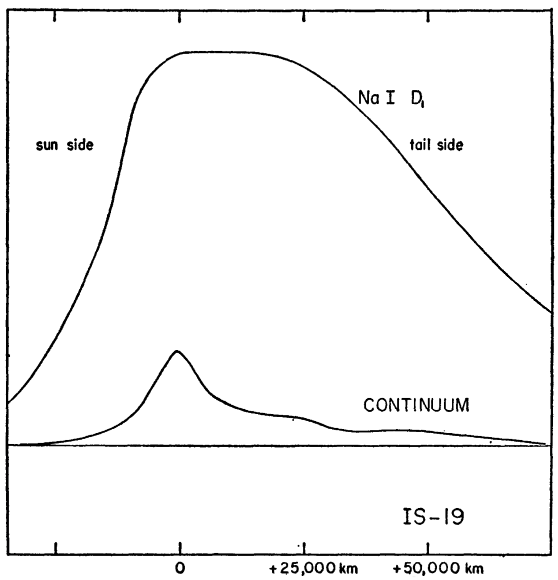

FIG. 9.—The spatial distribution of Na I emission (D_1-line) and dust (from the comet's reflected solar continuum at λ5893) on plate IS 19. Here the slit was along the Sun–comet-tail axis. Note the sharp nuclear condensation of dust giving a sharp peak in reflected light there, and the gradual decay of both Na I and dust well into the comet's tail.

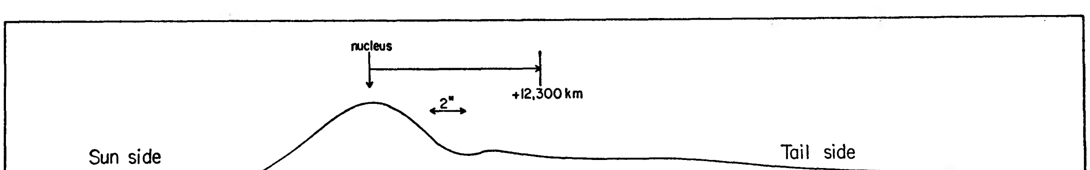

FIG. 10.—Details of the dust-continuum distribution on IS 19 near the nucleus of Comet Ikeya-Seki. The resolution of this plate is about 1″. Besides the dust observed in the tail, some also must be initially ejected toward the Sun to form the weak solar-side contribution; the dust particles will be gradually blown back toward the tail by solar radiation pressure. See the discussion in the text.

liberated neutral sodium atom is exposed to the full solar flux at $R = 8.6 R_{\odot}$. It is rapidly accelerated toward the tail, acquiring a velocity of about 18 km sec^{-1} away from the Sun. Then, after 50 sec, it is collisionally ionized or photo-ionized. The sodium ion is invisible to us and remains in the comet's tail, unless subjected to solar plasma charged-particle and/or magnetic-field effects (Biermann and Trefftz 1964; Biermann, Brosowski, and Schmidt 1967) not discussed here.

V. THE D-LINES SHADOW

One more feature we discuss from a rather phenomenological point of view is the noticeable "reversal" (see Fig. 11, Pl. 5); it can be also seen on tracing $y = -2, -6, -10$ on IS 20 (Fig. 7) as a small dip at the top of the profile at $v = +4$ km sec^{-1}. This velocity is quite constant along the whole reversal.

The reversal is not an optical depth or sodium-density effect as it is most noticeable in regions of average, not maximum, surrounding sodium emission-line intensity. The reversal is definitely directed away from the Sun; it is not seen at all at Sun-side positions on well-guided plates. For these two reasons, we suggest that the reversal is a real geometric shadow of the nucleus, both umbral and penumbral.

This interpretation implies that as seen by near-tail Na I atoms, the solar continuum is diluted by the shadow of many small particles and, possibly, a real asteroidal nucleus, whose dimensions are likely to be modest (cf. Roemer 1966). The cometary D-lines which are visible because of resonance scattering will then be weakened. The amount of weakening is not large (say ~20 per cent), but it is still easily noticeable. The only previous evidence of a similar phenomenon was the occultation of a star by Comet Burnham (1960 II) observed visually by Dossin (1962). The star was dimmed over 1 mag as the comet passed over it, crossing some 4″ from the comet's nucleus. The dusty nucleus of Comet 1965f might easily obscure the Sun slightly and cause a shadow in a similar fashion.

The maximum length of the shadow was estimated on several plates; the best guided plates with proper Sun–tail orientations yield a maximum length of ~12 mm or some 33000 km. The "nucleus" of the comet (the well-defined dust peak in Fig. 10) is about 2800 km in diameter, so here the shadow is about twelve times the diameter of the dust nucleus. We would expect a ratio of 4 to 1 for an isolated umbral shadow, so the penumbra must be adding considerably to the measured length, if our interpretation of the reversal is the correct one. The +4 km sec^{-1} velocity of the reversal is consistent with a shadow interpretation, as we expect to screen Na I atoms in the tail, and these must be moving away from the Sun to arrive in the comet's tail in the first place.

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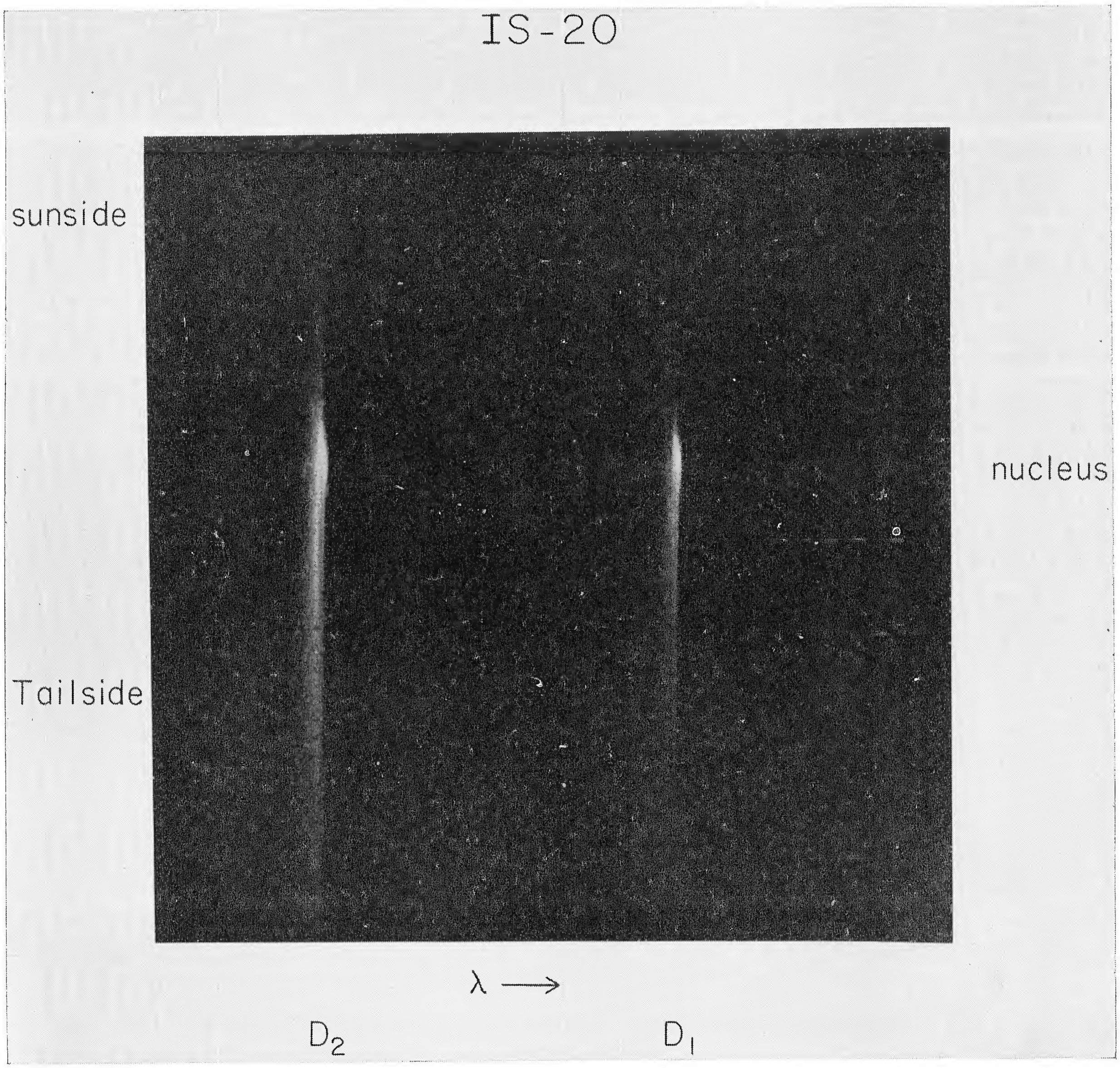

FIG. 11.—A dark print of IS 20 showing the D_2- and D_1-line "reversals," directed toward the tail of the comet. These features are interpreted as geometric shadows of the dusty nucleus of Comet 1965f.

SPINRAD AND MINER (*see* page 365)

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